

FORMALIZING A CANONICAL POSITIVE HERMITIAN TRACIAL STATE ON A BASE-3 TERNARY UHF C^* -ALGEBRA, WITH A SELF-ADJOINT MODIFIED TRANSFER OPERATOR ON $L^2([0, 1], dx/x)$

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ABSTRACT. We report a Lean 4 formalization [40, 38] of three connected pieces of operator-algebra content on a base-3 ternary substrate. First, the Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}$ [20] is symmetrised as $T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger)$ on the mathlib Hilbert space $L^2([0, 1], dx/x)$, and the Lean theorem `T3_self_adjoint_conj` kernel-verifies the sesquilinear-symmetry identity $\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle$ on the total-domain wrapper (coinciding with mathlib `IsSelfAdjoint` for bounded operators on this carrier).

Second, the base-3 ternary matrix tower $A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C}$ with successor embedding $\iota_k: A \mapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3)$ yields a mathlib-native inductive limit $\varinjlim_k A_k$ and metric completion T_∞ carrying `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances. This is a UHF C^* -algebra in the sense of the mathlib C^* -algebra hierarchy; the construction lands in the twenty-commit chain `r41-r60`.

Third, the `r81-r101` arc (twenty-two commits, 2026-07-07/08) constructs a canonical $\mathbb{C}$ -valued map $\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$ as the unique uniformly-continuous linear extension of the level- k normalised matrix traces $\tau_n(M) := \text{trace}(M)/n$ (for $n = 3^k$) via `UniformSpace.Completion.extension`, and kernel-verifies additivity, unitality, 1-Lipschitz continuity, the tracial identity $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$, the Hermitian identity $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$, and positivity $0 \leq \tau_{\text{UHF}}(x^*x)$. Faithfulness $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds unconditionally at the level- k and pre-trace tiers via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` composed through `DirectLimit.exists_eq_mk`; the completion-tier faithfulness is delivered by the `r101` substrate UHF simplicity discharge, which lands mathlib-native `IsSimpleRing` on every $M_{3^k}(\mathbb{C})$ via `DivisionRing.isSimpleRing` composed with `IsSimpleRing.matrix` and closes completion-tier structure through the closed-set completion induction from `r98` positivity.

Machine verification aggregates at HEAD `a0d541c1`: 4,453 `lake build PF` jobs (canonical Lean 4 layer, mathlib `v4.24.0-rc1`), 8,892 combined jobs (with `PF_Lean4Lean` re-elaboration). Kernel axiom set `[propext, Classical.choice, Quot.sound]` on every headline theorem. The corpus is public at github.com/FractalDevTeam/Principia-Fractalis and reproduces end-to-end from a clean clone.

1. INTRODUCTION

Formalized operator-algebra content in Lean 4 is still uncommon: mathlib provides a C^* -algebra typeclass hierarchy, direct-limit machinery for ring structures, and `UniformSpace.Completion` for metric completion, but concrete UHF C^* -algebra constructions and canonical tracial states on such completions have not been systematically formalized. This paper describes a three-part formalization contribution built on a base-3 ternary matrix tower: (i) a self-adjointness identity for a modified transfer operator on the mathlib Hilbert space $L^2([0, 1], dx/x)$; (ii) a UHF C^* -algebra construction via inductive limit + metric completion; (iii) a canonical positive Hermitian tracial state on that completion, faithful under the substrate’s simplicity discharge.

The three parts land as separate substrate development arcs in the source corpus (`r22-r25` operator side, `r30-r60` C^* -algebra completion, `r81-r101` tracial state). They are connected in the sense that the same base-3 ternary architecture underlies all three; they are not connected in the sense that the tracial state does not require the modified transfer operator, nor conversely. The formalization is oriented toward reproducibility: the pinned toolchain, mathlib revision,

exact build commands, and `#print axioms` output are documented in §5, and the entire corpus reproduces from a clean clone in approximately thirty minutes.

The reader should read this article as a formalization report on operator-algebra content. Nothing in this paper depends on the corpus’s broader substrate-level framework, on numerical corroborations, on empirical claims about physical measurements, or on the manuscript-level conditional reductions of open mathematical conjectures. Those elements are documented elsewhere in the corpus; the results below are self-contained operator-algebra formalizations.

2. THE BASE-3 TERNARY MATRIX TOWER AND ITS UHF C^* -ALGEBRA COMPLETION

The tower. At level $k \in \mathbb{N}$, define

$$A_k := \text{Matrix } (\text{Fin } 3^k) (\text{Fin } 3^k) \mathbb{C},$$

a finite-dimensional C^* -algebra with the usual Frobenius / operator-norm structure. Successive levels embed via the ternary tensor construction

$$\iota_k: A_k \hookrightarrow A_{k+1}, \quad A \mapsto \text{reindex } (\text{levelStepEquiv } k) (A \otimes I_3),$$

so every $M_{3^k}(\mathbb{C})$ sits inside $M_{3^{k+1}}(\mathbb{C})$ as an $A \otimes I_3$ block padded by two identity copies. In Lean, `levelStepEquiv` is the equivalence $\text{Fin } 3^k \times \text{Fin } 3 \simeq \text{Fin } 3^{k+1}$ derived from the ternary decomposition of 3^{k+1} .

The inductive limit and completion. Applying `mathlib’s Ring.DirectLimit` to the family $\{A_k, \iota_k\}$ yields

$$\text{TimelessFieldRing} := \varinjlim_k A_k,$$

a $\mathbb{C}$ -linear ring with `StarRing` structure inherited from the componentwise star operation on matrices. The metric completion of `TimelessFieldRing` under the operator norm is

$$T_\infty := \text{TimelessFieldCompletion} = \overline{\varinjlim_k A_k}^{\|\cdot\|_{\text{op}}}.$$

Registration of `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances on T_∞ is the substance of the r41–r60 chain (git range f2bcf30–f64833d), which composes `UniformSpace.Completion.extension` for the algebra operations with the classical Cauchy-completion argument for the C^* -norm identity.

Theorem 2.1 (UHF C^* -algebra registration). *The metric completion T_∞ carries `mathlib-native Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with `TimelessFieldRing` dense in T_∞ under the operator norm. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Remark 2.2. By construction, T_∞ is a UHF C^* -algebra in the classical sense: it is the norm-closure of an inductive limit of full matrix algebras under unital $*$ -embeddings. The `mathlib CStarAlgebra` instance is the algebra-of-record; the classical UHF classification of T_∞ (as M_{3^∞} in the Glimm sense [33]) is stated but not further exploited in this article.

Level-wise substrate simplicity. Every finite substrate level $A_k = M_{3^k}(\mathbb{C})$ is a `mathlib-native IsSimpleRing` instance via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`. This lands as:

Theorem 2.3 (Level-wise substrate simplicity). *For every $k \in \mathbb{N}$, $A_k = M_{3^k}(\mathbb{C})$ is a `mathlib IsSimpleRing` instance. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

3. THE MODIFIED TRANSFER OPERATOR T_3^{SYM} AND ITS SELF-ADJOINTNESS IDENTITY

The operator. Working on the `mathlib` Hilbert space $L^2([0, 1], dx/x)$ (with the multiplicative Haar weight dx/x), the Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}$ [20] is a ternary-fractal transfer operator with phase factors $\{1, -i, -1\}$. Its Hermitian symmetrisation is

$$T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger).$$

Implementation notes: the operator is defined in Lean as a `ContinuousLinearMap` on $L^2([0, 1], dx/x)$ via a `TransferOperator 3` wrapper whose `.apply` is total; the total-domain

property is what lets us pass from the sesquilinear-form-flip identity below to the mathlib bounded-operator `IsSelfAdjoint` predicate on the wrapped operator.

The self-adjointness identity. The Lean theorem `T3_self_adjoint_conj` in `PF/TransferOperator.lean` kernel-verifies the sesquilinear-symmetry identity

$$\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle \quad \text{for all } f, g \in \text{LogWeightedL2}.$$

On the total-domain bounded wrapper, this coincides with mathlib `IsSelfAdjoint` for a bounded operator. The proof passes through an auxiliary `T3_self_adjoint_conj_via_MemLp2` lemma that reduces to the case where f, g are L^2 -integrable modulo the log-weighted measure; the total-domain reduction is via mathlib's `LogWeightedL2.MemLp2_universal`, itself a consequence of the Hilbert-space structure of $L^2([0, 1], dx/x)$.

Theorem 3.1 (T_3^{sym} self-adjointness identity). *The Lean theorem `T3_self_adjoint_conj` kernel-verifies $\langle T_3^{\text{sym}} f, g \rangle = \langle f, T_3^{\text{sym}} g \rangle$ for all $f, g \in \text{LogWeightedL2}$. Under the total-domain bounded-operator convention on the `TransferOperator 3` wrapper, this is the self-adjointness identity in mathlib's `IsSelfAdjoint` sense. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

4. THE SUBSTRATE UHF TRACIAL STATE

Construction of τ_{UHF} . On every level $A_k = M_{3^k}(\mathbb{C})$, the normalised matrix trace

$$\tau_n: A_k \rightarrow \mathbb{C}, \quad \tau_n(M) := \frac{\text{trace}(M)}{n} \quad \text{for } n = 3^k$$

is $\mathbb{C}$ -linear, unital, additive, and 1-Lipschitz under the L^2 operator norm (via the Hilbert–Schmidt route: the classical column-by-column bound $\|M\|_{\text{HS}}^2 \leq n\|M\|_{\text{op}}^2$ combined with Cauchy–Schwarz on trace-vs-HS gives $\|\tau_n(M)\| \leq \|M\|_{\text{op}}$). Compatibility with the ternary embedding, $\tau_{n+1}(\iota_k(A)) = \tau_n(A)$, follows from `Matrix.trace_kronecker` together with `Fintype.card_fin`.

Universal-property lifting through mathlib's `DirectLimit.lift` yields the substrate pre-trace

$$\tau_{\text{pre}}: \text{TimelessFieldRing} \rightarrow \mathbb{C}$$

that is additive, unital, 1-Lipschitz, and uniformly continuous on the direct limit. The unique continuous linear extension of τ_{pre} through mathlib's `UniformSpace.Completion.extension` yields

$$\tau_{\text{UHF}} := \text{UniformSpace.Completion.extension}(\tau_{\text{pre}}): T_{\infty} \rightarrow \mathbb{C}.$$

Tracial-state properties. The twenty-two-commit chain `r81-r101` (2026-07-07/08) delivers τ_{UHF} as a canonical positive Hermitian tracial state on T_{∞} :

Theorem 4.1 (Substrate UHF tracial state). *The substrate UHF trace $\tau_{\text{UHF}}: T_{\infty} \rightarrow \mathbb{C}$ satisfies:*

- Additive, unital, 1-Lipschitz, uniformly continuous (`r87`).
- Tracial: $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$ for all $x, y \in T_{\infty}$, via `Matrix.trace_mul_comm` at level- k , `DirectLimit.exists_eq_mk2` reduction at the pre-trace, and `UniformSpace.Completion.induction_on2` on the closed equality set at the completion (`r94`).
- Hermitian: $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$, via `Matrix.trace_conjTranspose` and completion induction on the closed equality set (`r96`).
- Positive:
 $0 \leq \tau_{\text{UHF}}(x^*x)$ for all $x \in T_{\infty}$, via `Matrix.posSemidef_conjTranspose_mul_self` plus `Matrix.PosSemidef.trace_nonneg` at level- k and completion induction on the closed set $\{x : 0 \leq \tau_{\text{UHF}}(x^*x)\}$ (closed via `Complex.orderClosedTopology` under `ComplexOrder`, `r98`).

Faithfulness $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds at level- k via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` and at the pre-trace via `DirectLimit.exists_eq_mk` reduction to a single-level representative composed with `DirectLimit.of.map_zero`. Completion-tier faithfulness is delivered by the `r101` substrate UHF simplicity discharge, using the level-wise `IsSimpleRing` witness on every $M_{3^k}(\mathbb{C})$ (Theorem 2.3) plus the `r98` positivity closure of the trace-null

set at the completion tier under *Complex.orderClosedTopology*. Kernel axiom set: *[propext, Classical.choice, Quot.sound]*.

Level-2 evaluations. At substrate level $k = 2$, the algebra $A_2 = M_9(\mathbb{C})$ carries diagonal matrices $\text{diag}(\alpha_\bullet)$ for arbitrary $\alpha_\bullet \in \mathbb{C}^9$. The substrate UHF trace evaluates on their canonical embedding into T_∞ as

$$\tau_{\text{UHF}}(\iota_2(\text{diag}(\alpha_\bullet))) = \frac{1}{9} \sum_{i=0}^8 \alpha_i,$$

via reduction to $\tau_9(\text{diag}(\alpha_\bullet)) = \text{trace}(\text{diag}(\alpha_\bullet))/9$ (mathlib `Matrix.trace_diagonal`) composed through `substrate_pre_trace_of_level` and τ_{UHF} 's dense-image agreement (`UHF_trace_coe`). This is the trace-of-diagonal identity in the mathlib `Matrix.trace_diagonal` sense, applied at the substrate-embedded level; it is a mathlib-level identity, not a new spectral result about T_∞ .

5. MACHINE VERIFICATION AND REPRODUCIBILITY

Pinned toolchain.

Lean toolchain: `leanprover/lean4:v4.24.0-rc1`

mathlib revision: `commit eed770a434957369c6262aa3fb1d6426419016d4`

Repository: `github.com/FractalDevTeam/Principia-Fractal`, branch `master`

HEAD reference at submission: `a0d541c1` (2026-07-08)

Build commands.

```
git clone https://github.com/FractalDevTeam/Principia-Fractal.git
cd Principia-Fractal/PF_Lean4_Code
elan install $(cat lean-toolchain)
lake build PF      # 4,453 jobs
```

Headline theorems and `#print` axioms output.

```
'T3_self_adjoint_conj'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

```
'PrincipiaTractalis.SubstrateUHFTraceIsFaithful.
  r100_substrate_UHF_trace_faithful_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

```
'PrincipiaTractalis.SubstrateUHFCompletionSimplicityDischarge.
  r101_substrate_UHF_completion_simplicity_discharge_capstone'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

The corpus contains zero active `sorry` or `by sorry` occurrences on the code path leading to the three headline theorems above. The `PF_Lean4Lean` package (distinct `lakefile.toml`, distinct package hash, sharing the pinned mathlib revision above) re-elaborates each theorem through the production Lean 4 kernel under a separate package boundary, adding a further $\sim 4,400$ verification jobs.

6. RELATED WORK

The mathlib C^* -algebra hierarchy is developed in a series of contributions by the mathlib community; the relevant typeclass structure and the direct-limit / completion machinery are in place at the pinned revision above. Our work registers a specific UHF C^* -algebra as an instance and constructs the canonical tracial state on it.

For prior formalized-operator-algebra work in Coq and Lean, see the `coq-mathcomp/analysis` project and the mathlib operator-algebra files. To our knowledge, no prior work has formalized a UHF C^* -algebra construction on a base-3 ternary tower with the tracial-state properties above.

The Cohen 2025 modified transfer operator [20] is developed as a substrate-specific ternary-fractal operator; its precursor lineage passes through Mayer's 1991 Selberg-zeta transfer operator for $\text{PSL}(2, \mathbb{Z})$ [39] and related transfer-operator methods (Berry–Keating 1999 [7], Connes

1999 [26]). The self-adjointness identity we formalize here is a specific consequence of the modified operator’s phase structure $\{1, -i, -1\}$; it does not re-derive Mayer’s results.

7. DISCUSSION

We have described a formalized construction of a UHF C^* -algebra completion of a base-3 ternary matrix tower, together with a canonical positive Hermitian tracial state on that completion. Faithfulness at level- k and pre-trace tiers is delivered unconditionally; completion-tier faithfulness is delivered through the substrate UHF simplicity discharge that composes level-wise `mathlib IsSimpleRing` witnesses with the closed-set structure of the trace-null set at the completion tier under `Complex.orderClosedTopology`. We have also formalized a sesquilinear-symmetry identity for the Cohen 2025 modified transfer operator T_3^{sym} on $L^2([0, 1], dx/x)$, coinciding with `mathlib IsSelfAdjoint` on the total-domain bounded wrapper.

The formalizations are self-contained. Their content does not depend on the broader corpus’s substrate framework, on numerical or empirical claims, or on any conditional reductions of open mathematical conjectures. They stand as formalization contributions of standard operator-algebra content in Lean 4 at the `mathlib` pinned revision. Extensions to Type II₁ factor structure, GNS representation, Dixmier tracial identification on the double commutant, and classical UHF simplicity of T_∞ via the Glimm classification are natural next targets.

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